TH3 Task Design Document

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Overview

The Treasure Hunt 3 (TH3) task implements closed loop brain stimulation during a spatial memory task to test the hypothesis that direct electrical stimulation (DES) of the brain during poor memory states results in better spatial memory than during a control condition without brain stimulation. This work builds on several prior experiments: TH1 (record-only data during this task) and YC2 (open-loop brain stimulation during a different spatial memory task).

In TH1 we recorded brain activity from multiple implanted electrodes during memory encoding and retrieval in a task where patients were asked to remember the locations where particular objects were encountered. Our data analyses identified brain signals that correlated with whether object locations were correctly remembered (henceforth biomarkers). In order for an item's location to be considered correctly remembered, it must have met the following conditions. 1) The distance between the correct and the remembered location must be less than the median of all of the patient's distance errors (in cases where the median response is less than the radius of the selection circle, the radius of the selection circle will be used as a threshold instead), and 2) The patient must not have responded with the lowest confidence level.

We sought to predict whether a given object's location would be remembered based on the spectral decomposition of the time-series of bipolar neural recordings from all *N* channels recorded. We computed this prediction model by fitting a L2-regularized logistic regression to the data from each encoding event. In this model, the dependent variable was the recalled/non-recalled status of an object's location and the independent variables were the wavelet power in each of 8 frequency bands across all N channels (henceforth classifier). The model was fit to data from sessions 1 through M-1 and validated on session M, with the penalty parameter determined based on the distribution of best fitting parameters across all patients who performed the experiment (excluding the patient whose data were being fit). The model was able to reliably predict recall based on the spectral decomposition of the encoding period data with a mean classification area under the curve of .59.

The present study (TH3) aims to directly test the hypothesis that DES delivered during poor memory states can cause improvements in spatial memory compared with control conditions where no stimulation is applied. On half of the experiment lists, DES will be under control of an algorithm that evaluates the state of the brain for each presented object–location pair and decides whether or not to stimulate vs. control lists in which no stimulation is provided.

Methods

Except as noted, the design and procedure of the behavioral task is identical to TH1. The reader is referred to that design document for further details. Subjects will perform the Treasure Hunt task for 40 trials, divided into 5 blocks of eight. Stimulation is not delivered to the brain during lists 1-8 (Block 1) as these lists are used to evaluate the ability of the logistic-regression classifier, trained on the same patient's TH1 data, to predict the recalled status of items in TH3. Following the first 8 lists, the remaining 32 lists (Blocks 2-5) will be equally divided into two conditions: Stim and NoStim lists. These 32 lists will be pseudorandomly interleaved to balance the number of Stim and NoStim lists in the first and second halves of the session.

iEEG Analysis

During stimulation lists, intracranial EEG (iEEG) data are analyzed to determine when to turn on the stimulator. The raw iEEG data is converted from the default referential recording to a bipolar reference between neighboring electrodes as determined by the bipolar montage stored in each patient's '*Talairach structure*'[1]. These bipolar recordings are then filtered to remove 60Hz line noise using a Butterworth stop band filter (58-62Hz range, fourth order). A Morlet wavelet transformation is then applied to the signal (wave number = 5) to estimate spectral power during word encoding periods (8 wavelets log-spaced between 1-200Hz; center frequencies: [1.0, 2.1, 4.6, 9.7, 20.6, 44.0, 93.8, 200.0] Hz). For analysis of both the training (TH1) and closed-loop data (TH3) we use power at -1200–500 ms relative to the moment when each chest opens.

After applying the wavelet transformation we log-transform raw power values prior to removal of the buffer. These log transformed power values are then z-transformed to normalize them based on the within-session mean and standard deviation across all stimulus encoding events in the session. The resulting z-transformed log power values are then features for the pattern classifier described below.

<u>L2-logistic regression classifier:</u> Prior to the start of TH3 we will have already fit a regression model to data from multiple TH1 sessions, as described below. This fit of this regression model to ongoing iEEG data will determine the timing of stimulation in TH3 stim lists. Using the spectral features described above we will estimate an L2-penalized logistic regression model to predict whether an item's location will be subsequently recalled (a proxy for good memory). Before estimating the beta weights (coefficients) on the input features one must specify a penalty parameter to be used in the estimation routine. We determine the penalty parameter based on the penalty value that leads to the highest average AUC in all prior RAM subjects with at least two TH1 sessions. The penalty parameter will be inversely weighted according to the class imbalance (Hastie et al., 2001). Class weights for each class are computed as $(1/n_A)/((1/n_A + 1/n_B)/2)$ where n_A is the number of events class A events, and n_B is the number of class B events. This ensures best fit values of the cost parameter, C, are comparable across different class distributions. For example, if the proportion of observations in classes A/B are .25 and .75 respectively, then the ratio of the weights for the penalty parameters for each class will be 3:1. The weight for class A will be .25/((.25+.75)/2) = .5, and the weight for class B will be .75/(.25+.75)/2 = 1.5. As new participants are collected and analyzed, their best penalty parameter will be added to the distribution of all previous patients.

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[1] *Talairach structure* refers to the eeg_toolbox/PTSA data object used by the RAM team to represent information about each patient's montage. For the purposes of TH3, the relevant information is the set of bipolar electrode pairs, not the coordinate system used to represent the electrode spatial locations. The structure in fact contains electrode information in Montreal Neurlogical Institute (MNI) space; the 'Talairach' nomenclature has nonetheless been maintained from earlier versions of the toolboxes.

Criterion for progressing from TH1 to TH3. Each patient's classifier will be tested for out-of-sample performance using leave-one-session-out (LOSO) cross validation. The true LOSO AUC will be compared to an AUC distribution generated from classification of label-permuted input data. $P < 0.05$ one-tailed will be used as the significance threshold for continuing to TH3. At least three full task sessions are required. If classifier is not significant with three sessions, additional TH1 sessions should be collected.

Closed-loop Procedure

During the item presentation phase of the StimLists in TH3 we will evaluate the classifier output during the -1200–500-ms interval following the opening of each chest to decide whether to apply immediate stimulation. If the classifier output during this interval is below a threshold value we will stimulate immediately. We will use Youden's J statistic to determine the threshold for stimulation individually for each patient. This statistic is given by J = Sensitivity + Specificity – 1 and can be represented visually as the height of the ROC curve above the unity (chance) line. We select the maximal J value across all points in the patient's TH1 ROC curve as our threshold for stimulation (Schisterman et al., 2005). This is equivalent to the threshold that maximizes classifier performance for detecting true positives while minimizing false alarms.

Data Normalization Process. The values that are input to the real-time classifier will be normalized to ensure the range of the data match the range expected by the classifier model. This will be done using the mean and standard deviation across object–location encoding events in the first four word lists (one practice list and three NoStim lists). Mean and standard deviation estimates will be updated as more data are collected. Updates will occur after every NoStim list (following the recall period). To do so, StimControl.m will need state information about timing of word onset.

Classifier Validation. To validate the classifier for use in TH3, the output from the classifier for each item presented on the first four lists will be compared to the classifier output when tested on data from all previous sessions using a two-sample Kolmogorov-Smirnov test. The null hypothesis that the TH3 and training data come from the same distribution must not be rejected (i.e., $P > 0.05$) for the TH3 task to continue. If this hypothesis is rejected, an error will be thrown and the session will proceed as an TH1 session. StimControl.m will also output (to the command line and to a log file on disk) the results of the two-sample Kolmogorov-Smirnov test comparing the distributions.

Stimulation Onset Timing. A stimulation waveform, specified by the Parameter Search (PS) task (see below), will be triggered when the multivariate classifier detects that the classifier output is below threshold. Specifically, stimulation decisions will be made based on classifier decoding of the word presentation interval. If the classifier output for the decoded object falls below threshold, a stimulation waveform will be triggered.

Stimulation Artifacts. We will avoid decoding brain data during stimulation. We will accomplish this by decoding at the start of each object presentation and stimulating immediately after the 1700-ms word decoding interval (see *Stimulation Onset Timing* below). Because there are at least several seconds between the onsets of consecutive objects, there is no possibility of one stimulation affecting the subsequent decoding period.

Selection of Closed Loop Stimulation Parameters

We will use the *J* statistic from all of a given patient's TH1 sessions to determine threshold (*K*) for identifying poor memory states in all PS2 and PS3 task sessions (e.g., bottom 30th percentile, etc.). Across all such-identified poor memory states we will find the parameter set, $\vec{\theta}$ across all PS sessions that maximizes the t-statistic comparison between post-pre classifier output and the TH1 control comparison set. Based on the parameter manipulated across PS2 and PS3 tasks we expect to identify a set theta that includes:

- Stimulation location (bipolar electrode pair)
- Amplitude **(SafeAmp-.5, SafeAmp-.25 or SafeAmp mA)**
- Pulse frequency **(10, 25, 50, 100, 200 Hz or single pulse)**
- Theta pattern frequency **(3, 4, 5, 6, 7, 8 Hz or n/a)**
- Theta train frequency **(100, 200 Hz or n/a)**

In the absence of patient-specific PS data, $\vec{\theta}$ will be chosen according to the best group-average parameters from the vault of PS datasets in existence at the time of the patient's participation. *K* will be determined based on the patient's own TH1 data across all sessions using the J-statistic as above.

Other Stimulation Parameters:

- Based on the aggregate PS1 data pulse duration will be set to 500ms
- Bipolar stimulation
- Pulse width: 600 microsecond pulse (300 us per phase)

References

- Hastie. T., Tibishirani, R., & Friedman, J. (2001). The elements of statistical learning: data mining, inference and prediction. *New York: Springer.*
- Schisterman, E.F.; Perkins, N.J.; Liu, A.; Bondell, H. (2005). "Optimal cut-point and its corresponding Youden Index to discriminate individuals using pooled blood samples". *Epidemiology* **16**: 73–81.